

Problem set 5: International Trade II

C.Cooperation

1. Explain what is the “prisoner’s dilemma” and give on example (unrelated to prisoners) where the incentive structure of two parties is similar to a prisoner’s dilemma.
2. Explain how a bilateral international treaty that limits tariffs can be sustained, even if the choice of tariffs is similar to a prisoner’s dilemma.
3. John is studying for his exam in the course on “Economics & Law”. Jill is his neighbor and listens to loud music that disturbs John’s concentration. You are the landlord and must choose between two options:
 - a. Option 1: John can force Jill to stop the noise.
 - b. Option 2: Jill can keep on making noise.

From an efficiency perspective, does it matter which option you choose? What does it depend on? (in your answer, please refer to the Coase Theorem).

4. Alice declares herself the owner of the “deep water” in the ocean. Bob wants to spill chemicals in the water. He claims that the chemicals only stay in the “shallow water” and that Alice has no right to stop him. Alice claims that she not only has a right, but that she values her right by 1M\$. Bob claims that he values his right by 2M\$. Bob sues Alice. Alice says that the court should not intervene, because the Coase Theorem says that the parties should just negotiate and then they will reach an efficient outcome anyway. Is Alice’s correct?
5. Country A and Country B sign a free trade agreement in which they each obligate not to impose any tariffs. The two countries have the following payoff structure:

Country A / Country B	Tarif	No Tarif
Tarif	2, 2	10, 0
No Tarif	0, 10	7, 7

- a. Do the countries have a dominant strategy? If so, what is it?
- b. What is the (nash) equilibrium of the game if it is played one time?
- c. What is the (nash) equilibrium of the game if it is played 10 times?
- d. Suppose that both countries have the same discount factor, of $\delta = \frac{1}{2}$. What are then the (nash) equilibria of the game if it is played an *infinite amount of time* times?
- e. What is the critical discount factor for which the countries might prefer to cooperate by playing a grim trigger strategy (where each countries avoids charging a Tarif as long as the other country does so as well, and switches to charging a tariff forever once the other country charges a tariff).

Hint: you can use the formula discussed in class:

$$\delta \geq \frac{\Pi^D - \Pi^C}{\Pi^D - \Pi^P}$$

D. Classifying goods

6. Which one of these statements is *incorrect*?
 - a. A public good is non-rivalrous and non-excludible
 - b. A private good is rivalrous and excludible
 - c. A common good is non-rivalrous and excludible
 - d. A club good is non-rivalrous and excludible
7. Explain why private goods should be privately owned whereas public goods should be publicly owned, from an efficiency perspective.
8. Jane and John want to swim in the *Binnen Alster* (next to Jungfernstieg in Hamburg), but when they arrive there – they see that Max has sealed off the entire area. John files a petition to the court, saying Max should not be allowed to seal-off the area, because it is a public good. Is John correct?
9. Bill wants to buy a house that is owned by the government. The president objects to the sale, saying “This house is a public good, because it is owned by the government”. Is the president correct?
10. In the land of “Fantasialand”, the government owns the rights to the sky above the country. However, the neighboring countries disregard this right and fly airplanes above Fantasialand without approval. The president of Fantasialand is concerned that there is an incentive problem causing the government to under-invest in keeping the sky free of foreign planes. Hence, he decides to give the rights to Michael, a private person. Are the skies now a private good? If not, which type of good is it?
11. The king decides to build a wall around a small lake, which was previously operated in an “open-access” way (i.e. anyone could get in). Does this change the classification of the lake?